

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 2

24 JUNE 2025

Overview:

Today was the second day of our training. We learned about the **different types of Machine Learning** and got a chance to take a small test by ourselves to check our understanding. The session helped me to know how ML models are classified and how they work in real life.

Learning Objectives:

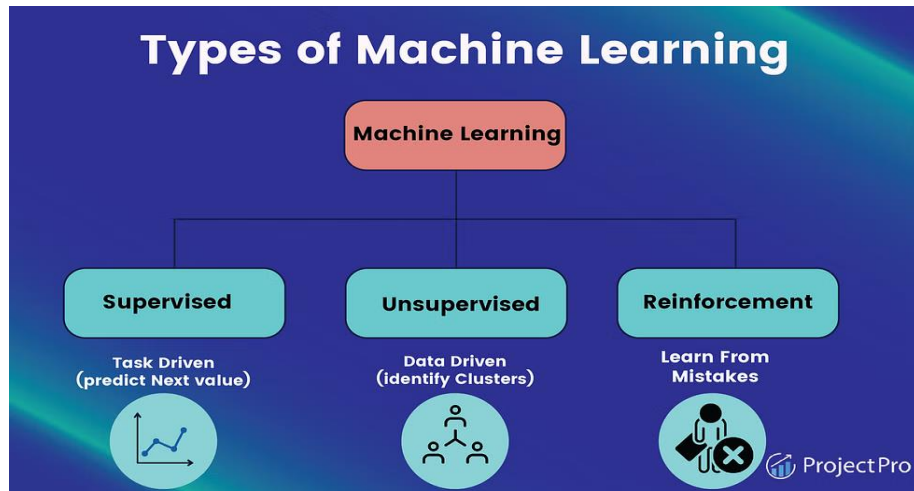
- Understand the main types of Machine Learning.
- Learn the difference between Supervised, Unsupervised, and Reinforcement Learning.
- Apply the knowledge through a self-test to check understanding.

Types of Machine Learning

There are several types of machine learning, each with special characteristics and applications. Some of the main types of machine learning algorithms are as follows:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Reinforcement Learning

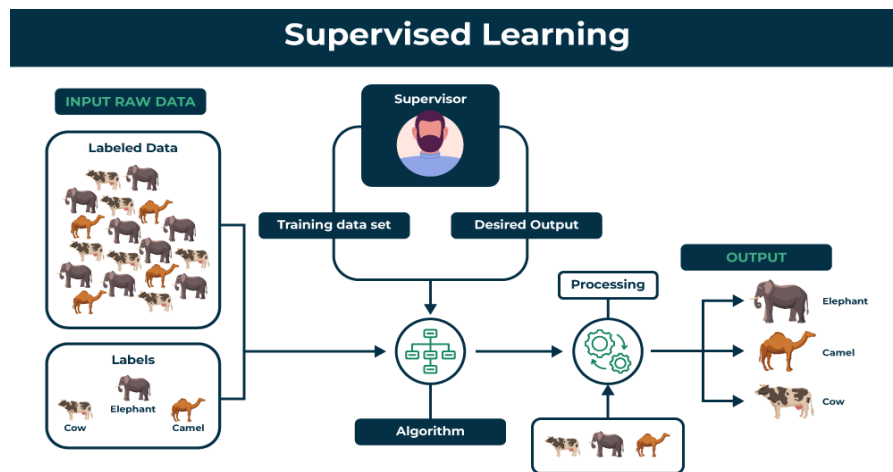
Also, there is a more specific category called semi-supervised learning, which combines elements of both supervised and unsupervised learning.



Types of Machine Learning

1. Supervised Machine Learning:

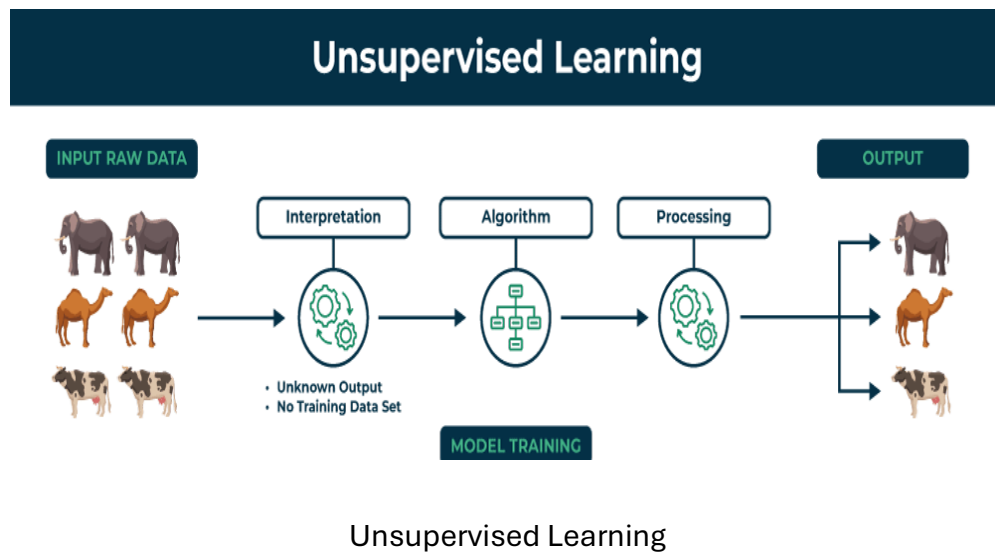
Supervised learning is defined as when a model gets trained on a Labelled Dataset. Labelled datasets have both input and output parameters. In Supervised Learning algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled



Supervised Learning

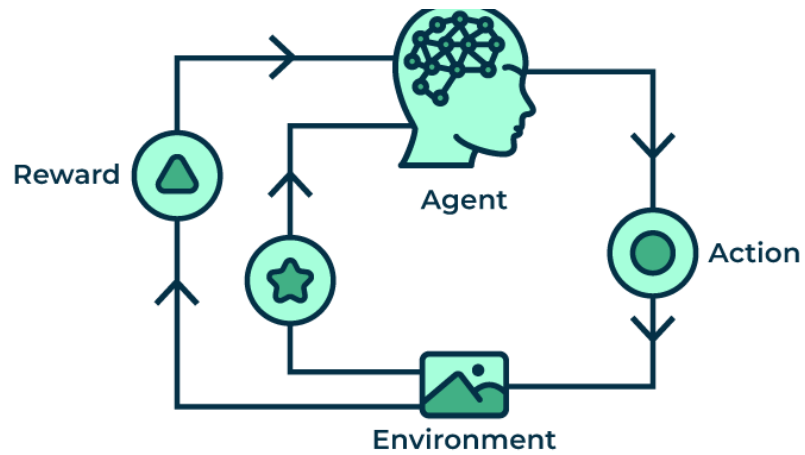
2. Unsupervised Machine Learning:

Unsupervised learning is a type of machine learning technique in which an algorithm discovers patterns and relationships using unlabeled data. Unlike supervised learning, unsupervised learning doesn't involve providing the algorithm with labeled target outputs. The goal of Unsupervised learning is often to discover hidden patterns, similarities, or clusters within the data, which can then be used for various purposes, such as data exploration, visualization, dimensionality reduction, and more.



3. Reinforcement Machine Learning:

Reinforcement machine learning algorithm is a learning method that interacts with the environment by producing actions and discovering errors. Trial, error, and delay are the most relevant characteristics of reinforcement learning. In this technique, the model keeps on increasing its performance using Reward Feedback to learn the behavior or pattern.



Reinforcement Machine Learning

Conclusion:

Learning about the types of Machines Learning made things a lot clearer. Understanding how supervised, unsupervised, and reinforcement learning work gave a better picture of how machines deal with different kinds of data. The self-test was helpful in checking how much I understood and where I need to improve. It was a good balance of theory and practice, and it made the topic more interesting and easier to remember.